# Solar Fe I appendix

Sun · Fe I · downloadable appendix

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|---|---|
| **Document** | Solar Fe I appendix |
| **Subject** | Solar photospheric abundance appendix · Fe I |
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Fe.json v1.531 · 183 Fe I products · 233 total

Solar Fe I anchor: 7.477 ± 0.037

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

## Highlighted band products

### near-UV

**7.452 ± 0.056**

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · n = 51

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

### VIS

**7.504 ± 0.034**

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

### VIS

**7.477 ± 0.037**

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

### red-optical

**7.440 ± 0.033**

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · n = 97

### NIR

**7.481 ± 0.034**

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 1

### H

**7.649 ± 0.042**

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · n = 8

## Error-bar forest

**How to read this forest.** Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Yellow rows are Deep Grade: strong lines that saturate, kept but less certain than the other grades. Unmeasured model rows remain N/A.

Grades and model labels

**Reference Grade.** An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

**Codex Grade.** The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.

**Deep Grade.** The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

**1D-LTE** uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

**3D-NLTE (Amarsi)** includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

**Mean-3D / Frankenstein** combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

**Synthesis** fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See Methodology for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

bar = σ as Asplund 2021 quote it (scatter, 3D, NLTE) · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders

### near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
### solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
### A(Fe) dex; axis 7.28–7.77

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Deep Grade · n=55 |  | 7.554<br>±0.068 |

### near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
### solar_kpno_kurucz2005_corrected
### A(Fe) dex; axis 7.28–7.77

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=48 |  | 7.515<br>±0.067 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=51 |  | 7.494<br>±0.058 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=48 |  | 7.523<br>±0.067 |

## near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.28–7.77

| | | |
|---|---|---|
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=51 |  | 7.559<br>±0.058 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=52 |  | 7.588<br>±0.062 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=53 |  | 7.678<br>±0.060 |

## near-UV — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
## solar_kpno_molecfit_corrected
## A(Fe) dex; axis 7.28–7.77

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=46 |  | 7.380<br>±0.071 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=50 |  | 7.398<br>±0.056 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=46 |  | 7.389<br>±0.071 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=51 |  | 7.452<br>±0.056 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=51 |  | 7.500<br>±0.058 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=51 |  | 7.543<br>±0.053 |

near-UV axis: 7.28 · 7.40 · 7.52 · 7.65 · 7.77 A(Fe) dex

## VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=22 |  | 7.443<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=22 |  | 7.450<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=18 |  | 7.462<br>±0.035 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=22 |  | 7.484<br>±0.035 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.531<br>±0.035 |

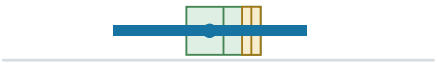
## VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.546<br>±0.036 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=21 |  | 7.471<br>±0.033 |

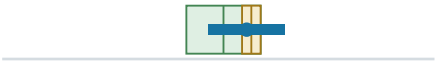
## VIS — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Deep Grade · n=106 |  | 7.329<br>±0.036 |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=101 |  | 7.340<br>±0.036 |

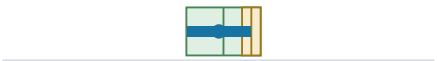
## VIS — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| EW · 1D-LTE<br>Codex Grade · n=13 · 1 alternate |  | 7.445<br>±0.104 |
| Synth · 1D-LTE<br>Codex Grade · n=65 |  | 7.410<br>±0.040 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=67 |  | 7.439<br>±0.067 |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=12 |  | 7.455<br>±0.107 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=59 |  | 7.411<br>±0.041 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=67 |  | 7.488<br>±0.066 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Codex Grade · n=67 |  | 7.541<br>±0.067 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Codex Grade · n=67 |  | 7.562<br>±0.066 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50 |  | 7.505<br>±0.043 |

## VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=166 |  | 7.367<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=167 |  | 7.373<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=156 |  | 7.371<br>±0.034 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=167 |  | 7.430<br>±0.034 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=101 |  | 7.485<br>±0.041 |

## VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=22 |  | 7.455<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=22 |  | 7.462<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=18 |  | 7.469<br>±0.036 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=22 |  | 7.495<br>±0.035 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.542<br>±0.035 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.562<br>±0.036 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=21 |  | 7.504<br>±0.034 |

## VIS — Kitt Peak — 1984 Molecfit corrected Codex deep pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Deep Grade · n=106 |  | 7.347<br>±0.036 |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=102 |  | 7.357<br>±0.037 |

## VIS — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| EW · 1D-LTE<br>Codex Grade · n=13 |  | 7.463<br>±0.111 |
| Synth · 1D-LTE<br>Codex Grade · n=65 |  | 7.414<br>±0.039 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=65 |  | 7.415<br>±0.038 |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=12 |  | 7.464<br>±0.112 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=59 |  | 7.420<br>±0.040 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=65 |  | 7.467<br>±0.038 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Codex Grade · n=65 |  | 7.511<br>±0.038 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Codex Grade · n=65 |  | 7.548<br>±0.038 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50 |  | 7.512<br>±0.043 |

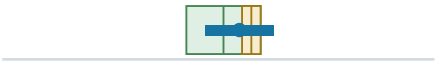
## VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=167 |  | 7.368<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=166 |  | 7.378<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=157 |  | 7.377<br>±0.034 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=167 |  | 7.432<br>±0.033 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=101 |  | 7.492<br>±0.041 |

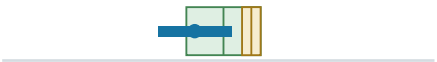
## VIS — HARPS Reference Grade · Asplund replication solar_harps_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=22 |  | 7.486<br>±0.040 |

## VIS — HARPS Reference Grade · Asplund replication solar_harps_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=22 |  | 7.491<br>±0.039 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=18 |  | 7.523<br>±0.042 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=22 |  | 7.532<br>±0.039 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.577<br>±0.041 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.617<br>±0.040 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=21 |  | 7.477<br>±0.037 |

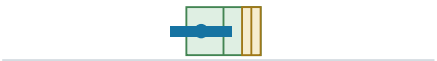
## VIS — HARPS Codex deep pool solar_harps_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Deep Grade · n=106 |  | 7.419<br>±0.039 |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=102 |  | 7.429<br>±0.040 |

## VIS — HARPS Codex deep pool · local continuum solar_harps_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Deep Grade · n=109 |  | 7.339<br>±0.087 |
| Synth · 1D-NLTE · Bergemann<br>Deep Grade · n=105 |  | 7.350<br>±0.088 |

## VIS — HARPS Codex graded pool solar_harps_molecfit_corrected A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| EW · 1D-LTE<br>Codex Grade · n=6 · 1 alternate |  | 7.498<br>±0.120 |
| Synth · 1D-LTE<br>Codex Grade · n=171 |  | 7.428<br>±0.033 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=171 |  | 7.436<br>±0.033 |
| EW · 1D-NLTE · Bergemann<br>Codex Grade · n=5 |  | 7.433<br>±0.123 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=144 |  | 7.429<br>±0.034 |

## VIS — HARPS Codex graded pool solar_harps_molecfit_corrected
## A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=171 |  | 7.492<br>±0.033 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Codex Grade · n=50 |  | 7.512<br>±0.043 |

## VIS — HARPS Reference Grade · full lab pool solar_harps_molecfit_corrected
## A(Fe) dex; axis 7.22–7.69

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=167 |  | 7.428<br>±0.036 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=167 |  | 7.441<br>±0.035 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=157 |  | 7.439<br>±0.036 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=167 |  | 7.491<br>±0.035 |
| Synth · 3D-NLTE · Amarsi · lab-gf<br>Reference Grade · n=100 |  | 7.500<br>±0.042 |

VIS axis: 7.22 · 7.34 · 7.45 · 7.57 · 7.69 A(Fe) dex

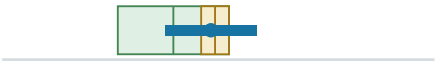
## red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=11 |  | 7.460<br>±0.044 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=11 |  | 7.464<br>±0.044 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=4 |  | 7.472<br>±0.046 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=11 |  | 7.507<br>±0.040 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.519<br>±0.045 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.562<br>±0.041 |

## red-optical — Kitt Peak — Kurucz 2005 corrected Codex graded pool solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=97 |  | 7.437<br>±0.033 |

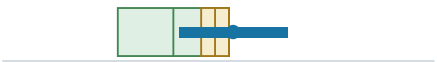
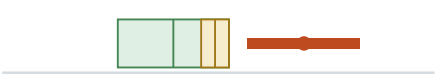
## red-optical — Kitt Peak — Kurucz 2005 corrected Codex graded pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=97 |  | 7.440<br>±0.033 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=46 |  | 7.449<br>±0.034 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=97 |  | 7.487<br>±0.033 |

## red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=68 |  | 7.423<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=66 |  | 7.421<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=34 |  | 7.418<br>±0.035 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=68 |  | 7.455<br>±0.034 |

## red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
## solar_kpno_molecfit_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=11 |  | 7.449<br>±0.043 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=11 |  | 7.451<br>±0.043 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=4 |  | 7.488<br>±0.048 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=11 |  | 7.503<br>±0.039 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.525<br>±0.044 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.554<br>±0.041 |

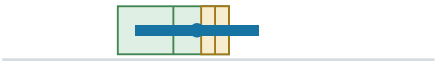
## red-optical — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=66 |  | 7.413<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=67 |  | 7.419<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=33 |  | 7.403<br>±0.036 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=65 |  | 7.454<br>±0.035 |

## red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=66 |  | 7.413<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=67 |  | 7.419<br>±0.034 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=33 |  | 7.403<br>±0.036 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=65 |  | 7.454<br>±0.035 |

## red-optical — IAG Reference Grade · Asplund replication solar_iag
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=11 |  | 7.477<br>±0.044 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=11 |  | 7.480<br>±0.044 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=4 |  | 7.505<br>±0.050 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=11 |  | 7.522<br>±0.041 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.545<br>±0.045 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=11 |  | 7.576<br>±0.042 |

## red-optical — IAG Codex graded pool solar_iag
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=96 |  | 7.437<br>±0.036 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=96 |  | 7.437<br>±0.036 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=46 |  | 7.446<br>±0.045 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=95 |  | 7.479<br>±0.036 |

## red-optical — IAG Reference Grade · full lab pool solar_iag
## A(Fe) dex; axis 7.34–7.65

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=67 |  | 7.424<br>±0.039 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=66 |  | 7.423<br>±0.039 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=33 |  | 7.422<br>±0.052 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=66 |  | 7.454<br>±0.039 |

red-optical axis: 7.34 · 7.41 · 7.49 · 7.57 · 7.65 A(Fe) dex

## NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=1 |  | 7.478<br>±0.034 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=1 |  | 7.481<br>±0.034 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=1 |  | 7.525<br>±0.034 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.533<br>±0.035 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.569<br>±0.035 |

## NIR — Kitt Peak — Kurucz 2005 corrected Codex graded pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=21 |  | 7.517<br>±0.055 |

## NIR — Kitt Peak — Kurucz 2005 corrected Codex graded pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=21 |  | 7.508<br>±0.055 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=5 |  | 7.538<br>±0.093 |

## NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
## solar_kpno_kurucz2005_corrected
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=21 |  | 7.517<br>±0.055 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=21 |  | 7.500<br>±0.055 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=5 |  | 7.538<br>±0.093 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=19 |  | 7.552<br>±0.055 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=22 |  | 7.586<br>±0.055 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=20 |  | 7.622<br>±0.054 |

## NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
## solar_kpno_molecfit_corrected
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=1 |  | 7.451<br>±0.043 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=1 |  | 7.454<br>±0.041 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=1 |  | 7.498<br>±0.041 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.506<br>±0.041 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.541<br>±0.041 |

## NIR — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=18 |  | 7.495<br>±0.047 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=20 |  | 7.482<br>±0.045 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=5 |  | 7.368<br>±0.068 |

## NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=18 |  | 7.495<br>±0.047 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=20 |  | 7.478<br>±0.059 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=5 |  | 7.368<br>±0.068 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=20 |  | 7.514<br>±0.060 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=20 |  | 7.559<br>±0.059 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=20 |  | 7.577<br>±0.060 |

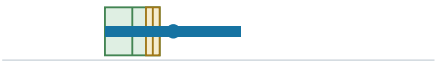
## NIR — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=1 |  | 7.467<br>±0.035 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=1 |  | 7.471<br>±0.035 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=1 |  | 7.514<br>±0.035 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.523<br>±0.036 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=1 |  | 7.558<br>±0.036 |

## NIR — IAG Codex graded pool solar_iag
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=24 |  | 7.460<br>±0.075 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=23 |  | 7.468<br>±0.078 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=7 |  | 7.369<br>±0.069 |

## NIR — IAG Reference Grade · full lab pool solar_iag
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=17 |  | 7.520<br>±0.099 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=17 |  | 7.493<br>±0.099 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=4 |  | 7.515<br>±0.112 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=17 |  | 7.555<br>±0.099 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=17 |  | 7.548<br>±0.098 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=17 |  | 7.598<br>±0.098 |

## NIR — CRIRES+ Codex graded pool solar_crires_plus_y_wide_rya1054
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=5 |  | 7.551<br>±0.093 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=5 |  | 7.552<br>±0.092 |
| Synth · 1D-NLTE · Bergemann<br>Codex Grade · n=2 |  | 7.637<br>±0.235 |

## NIR — CRIRES+ Reference Grade · full lab pool solar_crires_plus_y_wide_rya1054
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=5 |  | 7.551<br>±0.093 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=5 |  | 7.557<br>±0.085 |
| Synth · 1D-NLTE · Bergemann<br>Reference Grade · n=2 |  | 7.637<br>±0.235 |

## NIR — CRIRES+ Reference Grade · full lab pool solar_crires_plus_y_wide_rya1054
## A(Fe) dex; axis 7.27–7.90

| | | |
|---|---|---|
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=5 |  | 7.594<br>±0.096 |
| Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=5 |  | 7.607<br>±0.080 |
| Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental)<br>Reference Grade · n=5 |  | 7.635<br>±0.094 |

NIR axis: 7.27 · 7.43 · 7.59 · 7.74 · 7.90 A(Fe) dex

## H — CRIRES+ Codex graded pool solar_crires_plus_h_rya1232
## A(Fe) dex; axis 7.39–7.72

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Codex Grade · n=19 |  | 7.497<br>±0.046 |
| Synth · 1D-LTE · Gerber<br>Codex Grade · n=19 |  | 7.502<br>±0.046 |
| Synth · 1D-NLTE · Gerber<br>Codex Grade · n=8 |  | 7.649<br>±0.042 |

## H — CRIRES+ Reference Grade · full lab pool solar_crires_plus_h_rya1232
## A(Fe) dex; axis 7.39–7.72

| | | |
|---|---|---|
| Synth · 1D-LTE<br>Reference Grade · n=19 |  | 7.497<br>±0.046 |
| Synth · 1D-LTE · Gerber<br>Reference Grade · n=19 |  | 7.502<br>±0.046 |
| Synth · 1D-NLTE · Gerber<br>Reference Grade · n=8 |  | 7.649<br>±0.042 |

H axis: 7.39 · 7.47 · 7.56 · 7.64 · 7.72 A(Fe) dex

Read all product values, including alternate line sets and experiments

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · 1D-LTE: 7.497 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.042 (Asplund 2021 convention); n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · 1D-LTE: 7.497 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.042 (Asplund 2021 convention); n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · 1D-LTE: 7.551 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · ENGINE-A: 7.637 ± 0.235 (Asplund 2021 convention); n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.552 ± 0.092 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · 1D-LTE: 7.551 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · ENGINE-A: 7.637 ± 0.235 (Asplund 2021 convention); n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.594 ± 0.096 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.557 ± 0.085 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.607 ± 0.080 (Asplund 2021 convention); n=5; ξ MEASURED · experimental, not adopted

CRIRES+ · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.635 ± 0.094 (Asplund 2021 convention); n=5; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.467 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.514 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.471 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.523 ± 0.036 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.558 ± 0.036 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.460 ± 0.075 (Asplund 2021 convention); n=24; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.369 ± 0.069 (Asplund 2021 convention); n=7; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.468 ± 0.078 (Asplund 2021 convention); n=23; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.520 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.515 ± 0.112 (Asplund 2021 convention); n=4; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.555 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.493 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.548 ± 0.098 (Asplund 2021 convention); n=17; ξ MEASURED · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.598 ± 0.098 (Asplund 2021 convention); n=17; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.478 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.525 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.481 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.533 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.569 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.517 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.538 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.508 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.517 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.538 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.552 ± 0.055 (Asplund 2021 convention); n=19; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.500 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.586 ± 0.055 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.622 ± 0.054 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.451 ± 0.043 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.498 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.454 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.506 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.541 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.495 ± 0.047 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.368 ± 0.068 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.482 ± 0.045 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.495 ± 0.047 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.368 ± 0.068 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.514 ± 0.060 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.478 ± 0.059 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.559 ± 0.059 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.577 ± 0.060 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.477 ± 0.037 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.486 ± 0.040 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.523 ± 0.042 (Asplund 2021 convention); n=18; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.532 ± 0.039 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.491 ± 0.039 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.577 ± 0.041 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

HARPS · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.617 ± 0.040 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · 1D-LTE: 7.339 ± 0.087 (Asplund 2021 convention); n=109; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · ENGINE-A: 7.350 ± 0.088 (Asplund 2021 convention); n=105; ξ MEASURED

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.419 ± 0.039 (Asplund 2021 convention); n=106; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.429 ± 0.040 (Asplund 2021 convention); n=102; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.120 (Asplund 2021 convention); n=6; ξ MEASURED

HARPS · EW · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.433 ± 0.123 (Asplund 2021 convention); n=5; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.431 ± 0.122 (Asplund 2021 convention); n=5; ξ MEASURED

HARPS · Synth · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.428 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.429 ± 0.034 (Asplund 2021 convention); n=144; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.492 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.436 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.428 ± 0.036 (Asplund 2021 convention); n=167; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.500 ± 0.042 (Asplund 2021 convention); n=100; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.439 ± 0.036 (Asplund 2021 convention); n=157; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.491 ± 0.035 (Asplund 2021 convention); n=167; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.441 ± 0.035 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.471 ± 0.033 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.443 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.462 ± 0.035 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.484 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.450 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.531 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.546 ± 0.036 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.329 ± 0.036 (Asplund 2021 convention); n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.340 ± 0.036 (Asplund 2021 convention); n=101; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.445 ± 0.104 (Asplund 2021 convention); n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.455 ± 0.107 (Asplund 2021 convention); n=12; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.479 ± 0.090 (Asplund 2021 convention); n=9; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.410 ± 0.040 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.505 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.411 ± 0.041 (Asplund 2021 convention); n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.488 ± 0.066 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.439 ± 0.067 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.541 ± 0.067 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.066 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.367 ± 0.034 (Asplund 2021 convention); n=166; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.485 ± 0.041 (Asplund 2021 convention); n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.371 ± 0.034 (Asplund 2021 convention); n=156; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.430 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.373 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.034 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.455 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.469 ± 0.036 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.495 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.462 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.542 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.036 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.347 ± 0.036 (Asplund 2021 convention); n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.357 ± 0.037 (Asplund 2021 convention); n=102; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.463 ± 0.111 (Asplund 2021 convention); n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.464 ± 0.112 (Asplund 2021 convention); n=12; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.414 ± 0.039 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.420 ± 0.040 (Asplund 2021 convention); n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.467 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.415 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.511 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.548 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.368 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.492 ± 0.041 (Asplund 2021 convention); n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.377 ± 0.034 (Asplund 2021 convention); n=157; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.432 ± 0.033 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.378 ± 0.034 (Asplund 2021 convention); n=166; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.554 ± 0.068 (Asplund 2021 convention); n=55; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.515 ± 0.067 (Asplund 2021 convention); n=48; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.523 ± 0.067 (Asplund 2021 convention); n=48; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.559 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.494 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.588 ± 0.062 (Asplund 2021 convention); n=52; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.678 ± 0.060 (Asplund 2021 convention); n=53; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.380 ± 0.071 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.389 ± 0.071 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.452 ± 0.056 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.398 ± 0.056 (Asplund 2021 convention); n=50; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.500 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.543 ± 0.053 (Asplund 2021 convention); n=51; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.477 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.505 ± 0.050 (Asplund 2021 convention); n=4; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.522 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.480 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.545 ± 0.045 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.576 ± 0.042 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.036 (Asplund 2021 convention); n=96; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.446 ± 0.045 (Asplund 2021 convention); n=46; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.479 ± 0.036 (Asplund 2021 convention); n=95; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.437 ± 0.036 (Asplund 2021 convention); n=96; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.424 ± 0.039 (Asplund 2021 convention); n=67; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.422 ± 0.052 (Asplund 2021 convention); n=33; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.039 (Asplund 2021 convention); n=66; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.423 ± 0.039 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.460 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.472 ± 0.046 (Asplund 2021 convention); n=4; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.507 ± 0.040 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.464 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.519 ± 0.045 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.449 ± 0.034 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.487 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.440 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.423 ± 0.034 (Asplund 2021 convention); n=68; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.418 ± 0.035 (Asplund 2021 convention); n=34; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.455 ± 0.034 (Asplund 2021 convention); n=68; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.421 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.449 ± 0.043 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.488 ± 0.048 (Asplund 2021 convention); n=4; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.503 ± 0.039 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.451 ± 0.043 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.525 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.554 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.413 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.403 ± 0.036 (Asplund 2021 convention); n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.035 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.034 (Asplund 2021 convention); n=67; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.413 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.403 ± 0.036 (Asplund 2021 convention); n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.035 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.034 (Asplund 2021 convention); n=67; ξ MEASURED

## Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

## Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

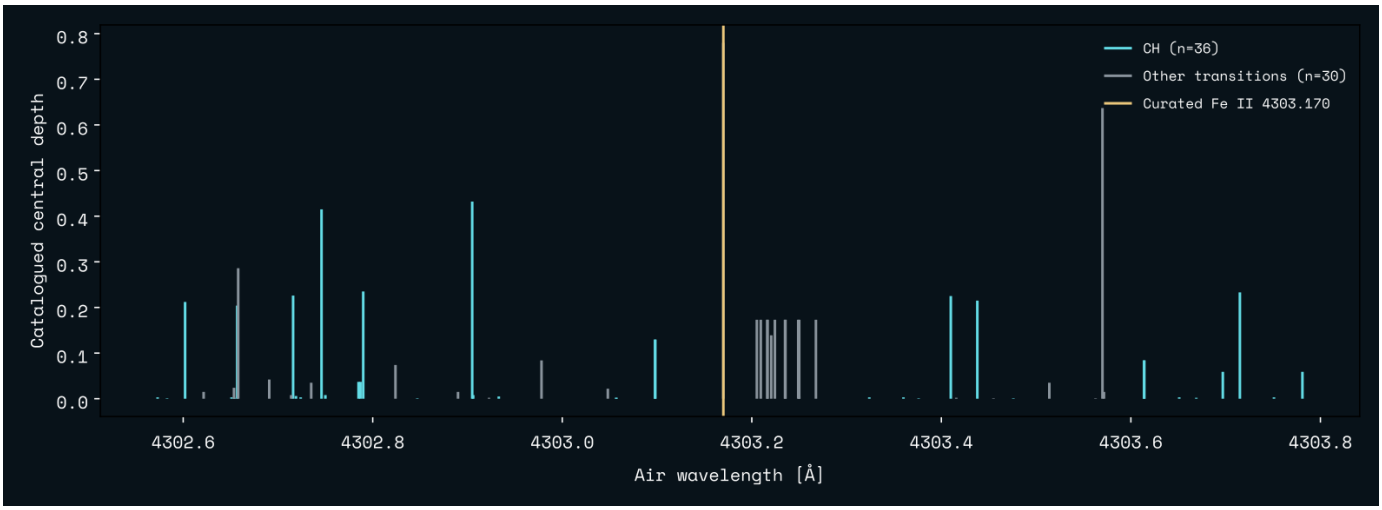
## Frankenstein — experimental, pending the Bride

42 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

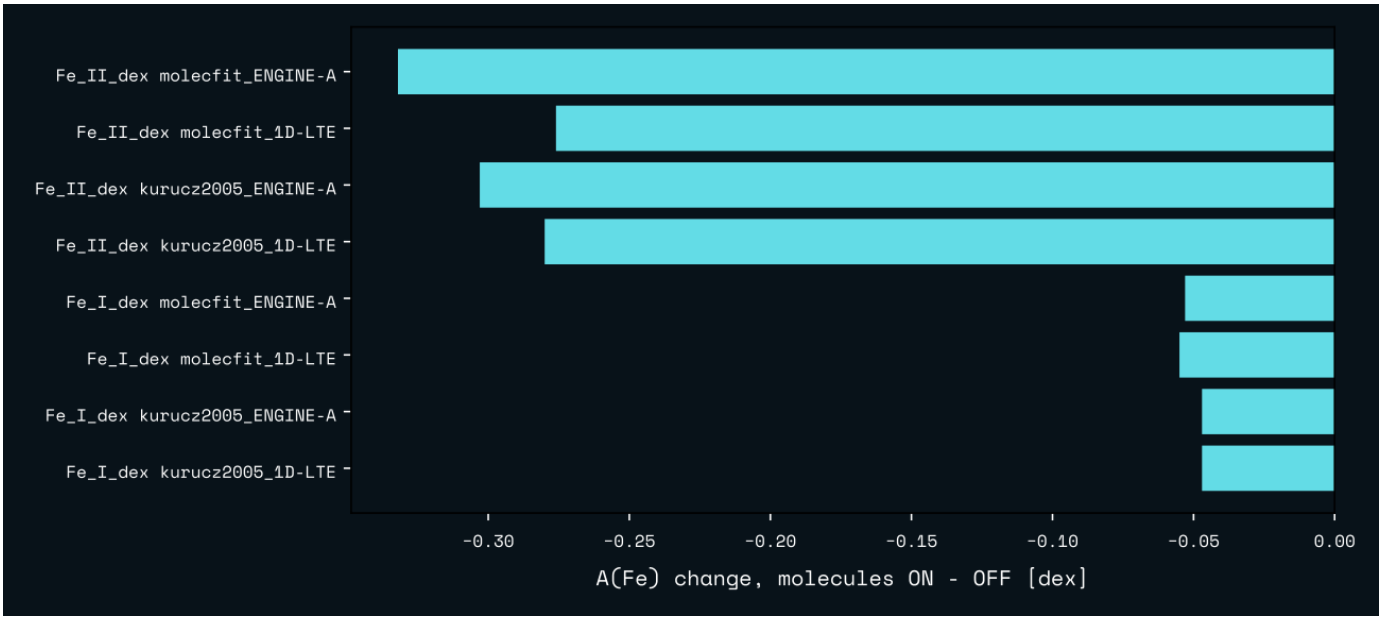
## Problem-line and model diagnostics

### CH G-band artifact



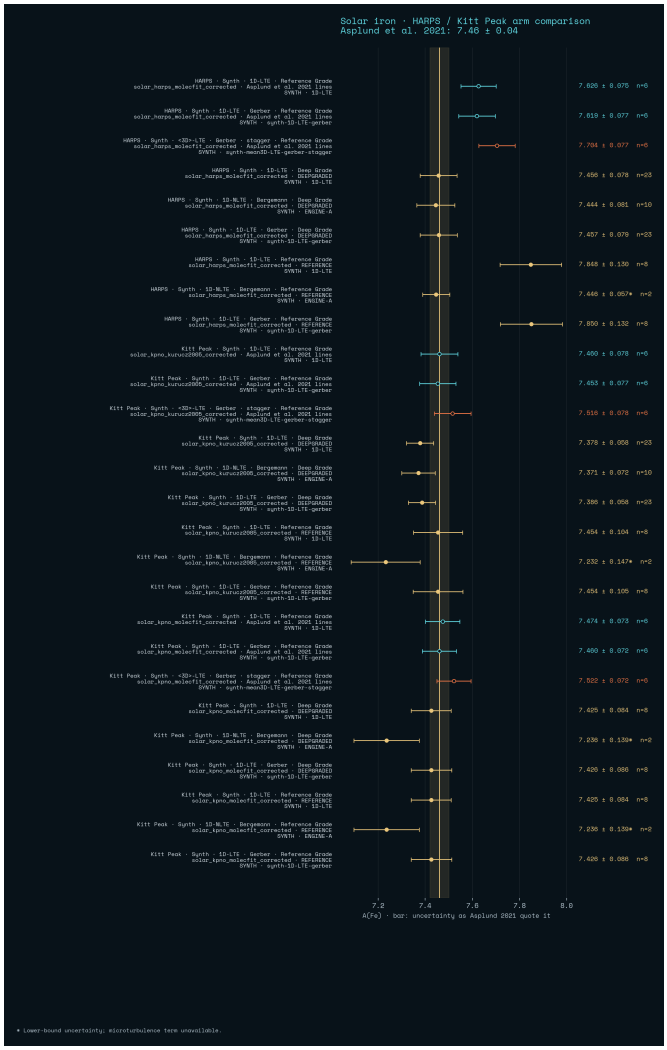

CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. Source evidence

### Near-UV molecular opacity



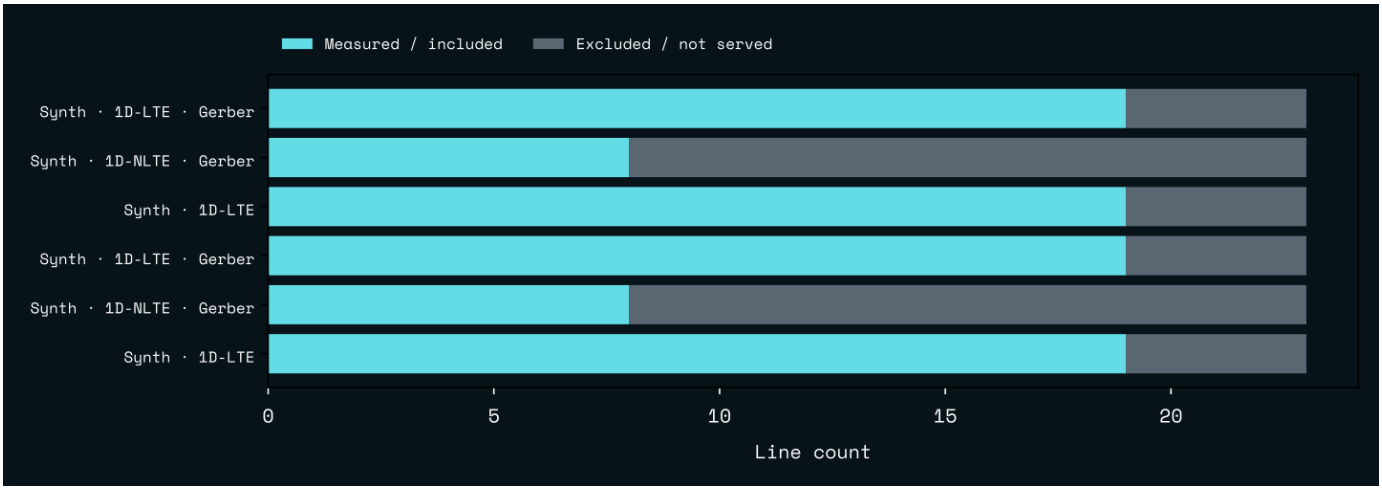

molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

## HARPS arm offset




Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

## CRIRES+ H model coverage




Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

## Download the evidence

692 graded per-line measurements exported. Evidence for 149 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

## Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit 425672429b119fd8bd38b40991c0b2685635f414
Fe.json v1.531 · feed timestamp 2026-10-29T03:00:31Z
Generated 2026-10-29T03:25:04+00:00
Feed SHA-256 668258fe7ac5b9f4fb0836eb00d9fc857c0ed693d589563e2e7c586b76f54fc2

Refreshed product tracker · Build manifest · Fe II appendix

## References

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Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Amarsi, A. M.; Liljegren, S.; Nissen, P. E. (2022). 3D non-LTE iron abundances in FG-type dwarfs (doi: 10.1051/0004-6361/202244542). The Amarsi 3D non-LTE iron correction grid used by the displayed products.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Ruffoni, M. P.; Lawler, J. E.; Pickering, J. C.; Lind, K.; Brewer, N. R. (2014). Fe I Oscillator Strengths for Transitions from High-lying Even-parity Levels (doi: 10.1088/0067-0049/215/2/23). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). Fe I oscillator strengths for the Gaia-ESO survey (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.